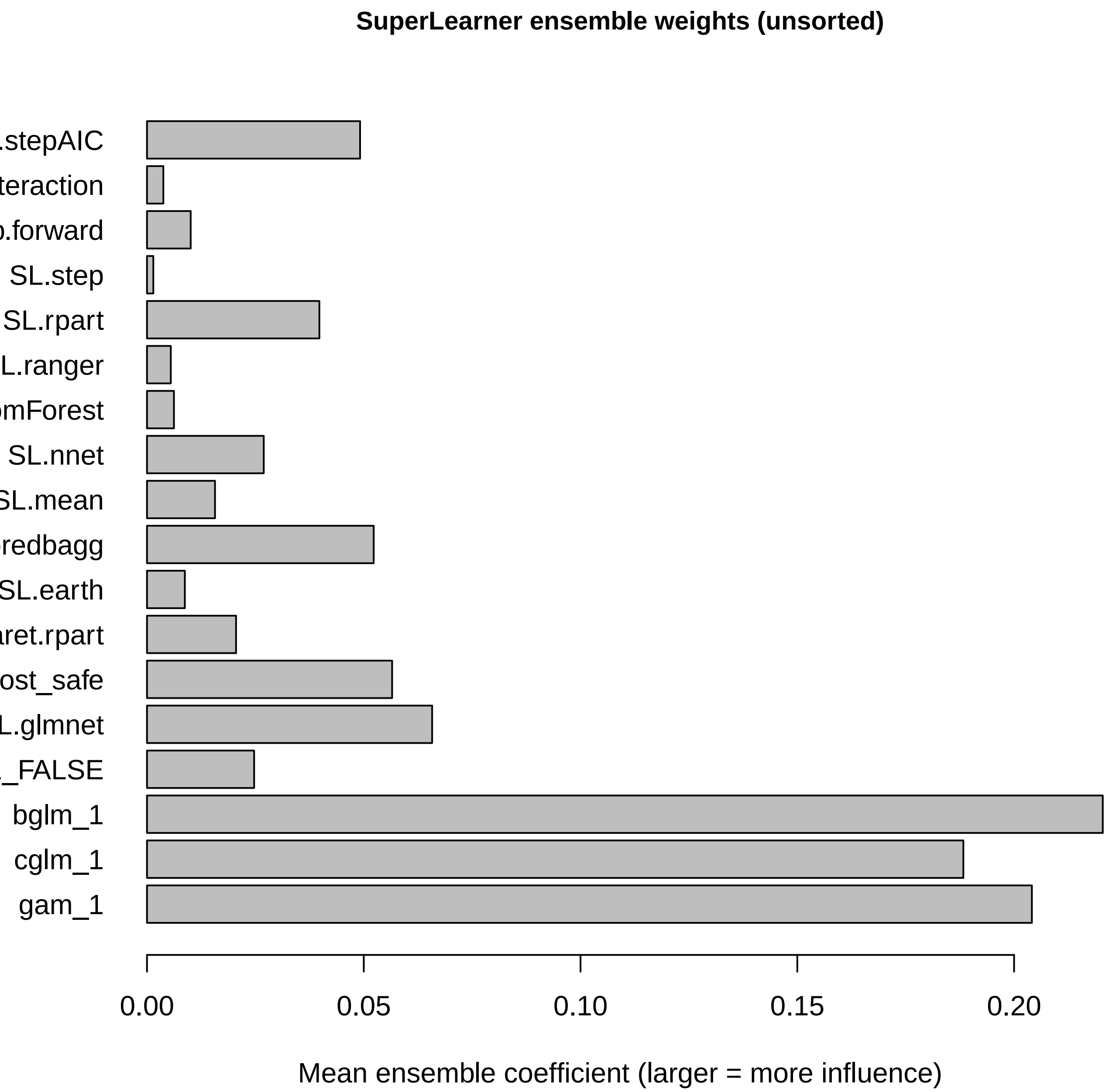
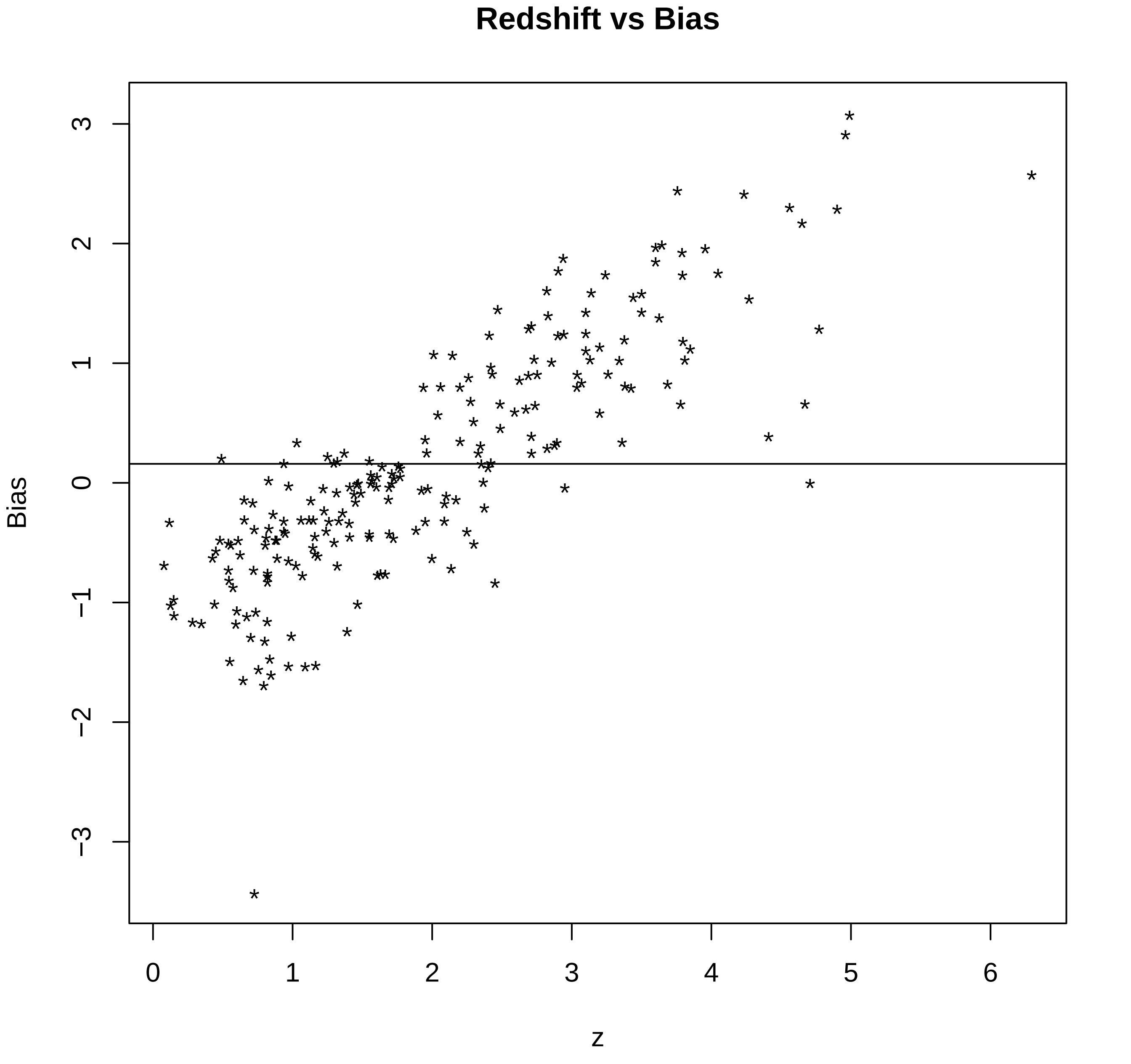
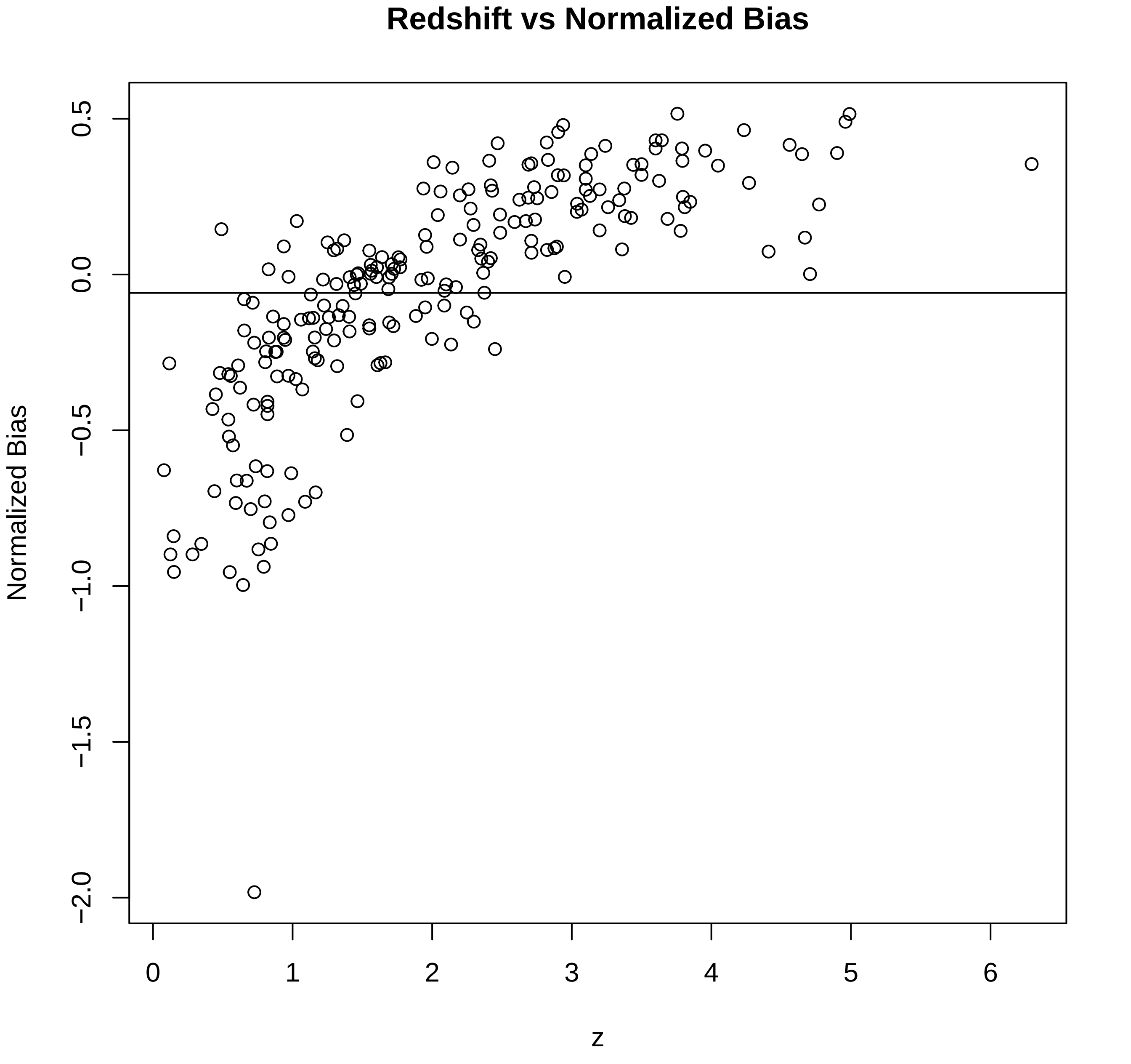
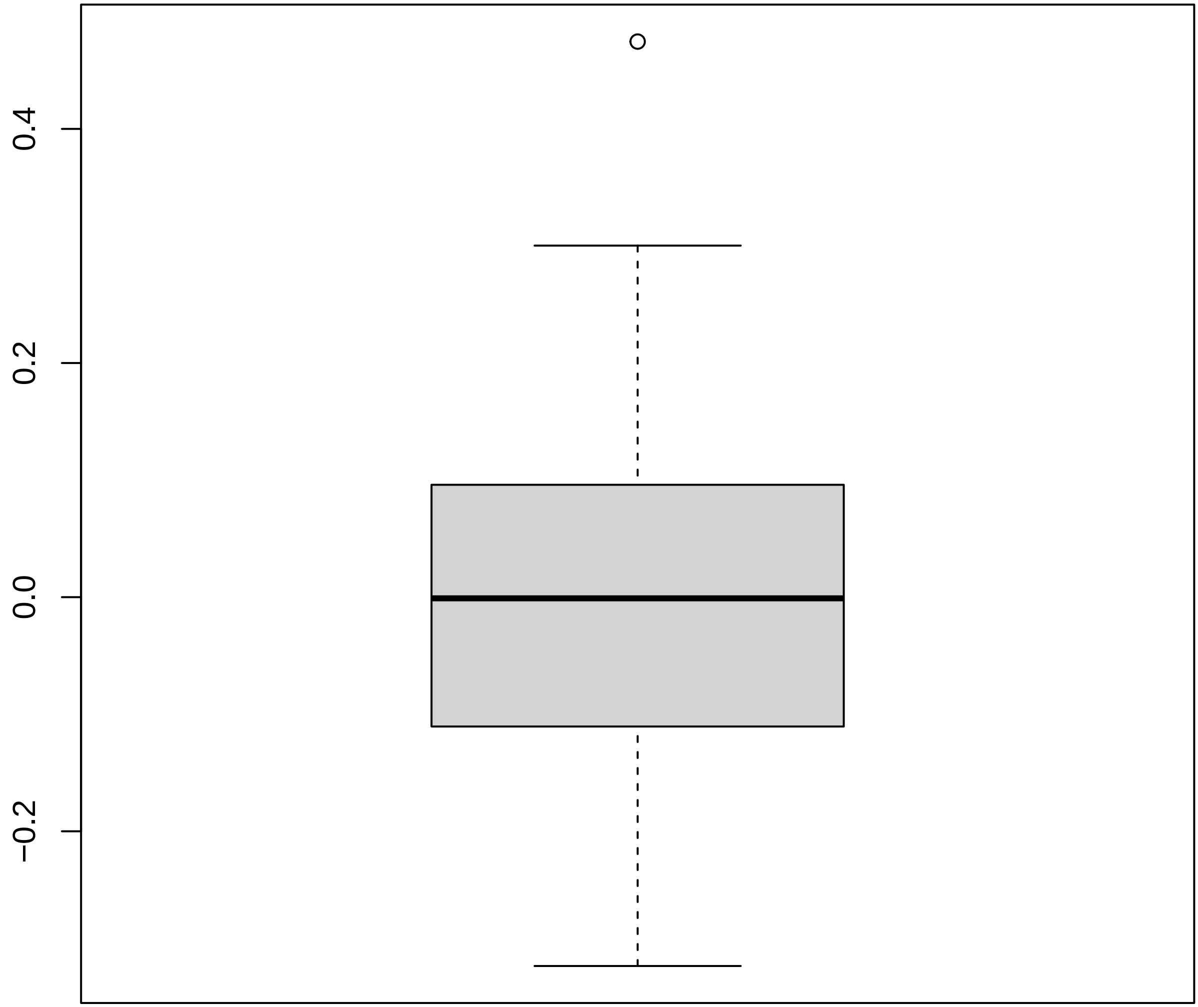

SuperLearner ensemble weights (unsorted)
.stepAIC
teraction
o.forward
SL.step
SL.rpart
L.ranger
omForest
SL.nnet
SL.mean
redbagg
SL.earth
aret.rpart
ost_safe
L.glmnet
_FALSE
bglm_1
cglm_1
gam_1
0.00
0.05
0.10
0.15
0.20
Mean ensemble coefficient (larger = more influence)

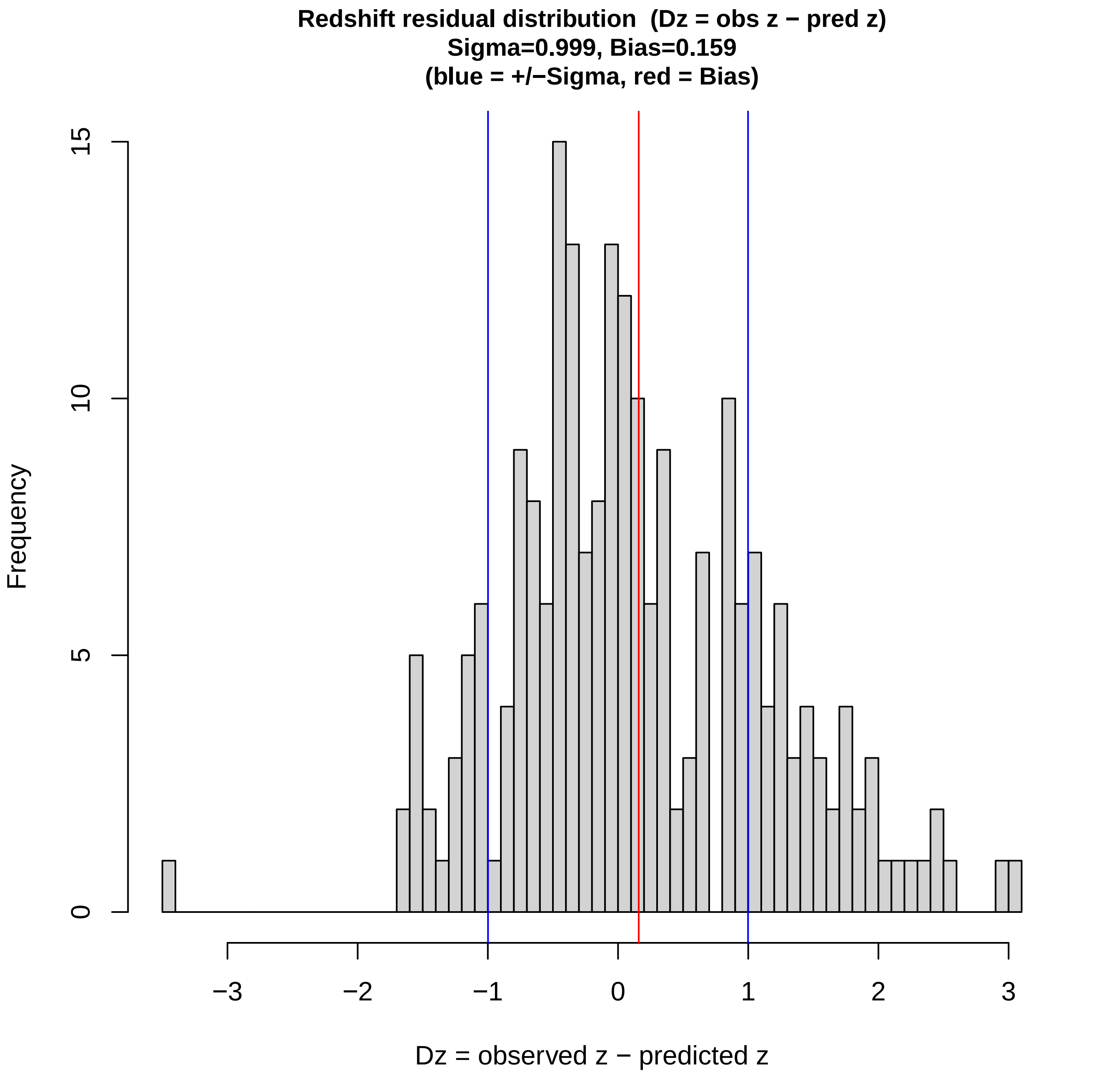


Redshift vs Bias
Bias
z
3
2
1
0
−1
−2
−3
0
1
2
3
4
5
6

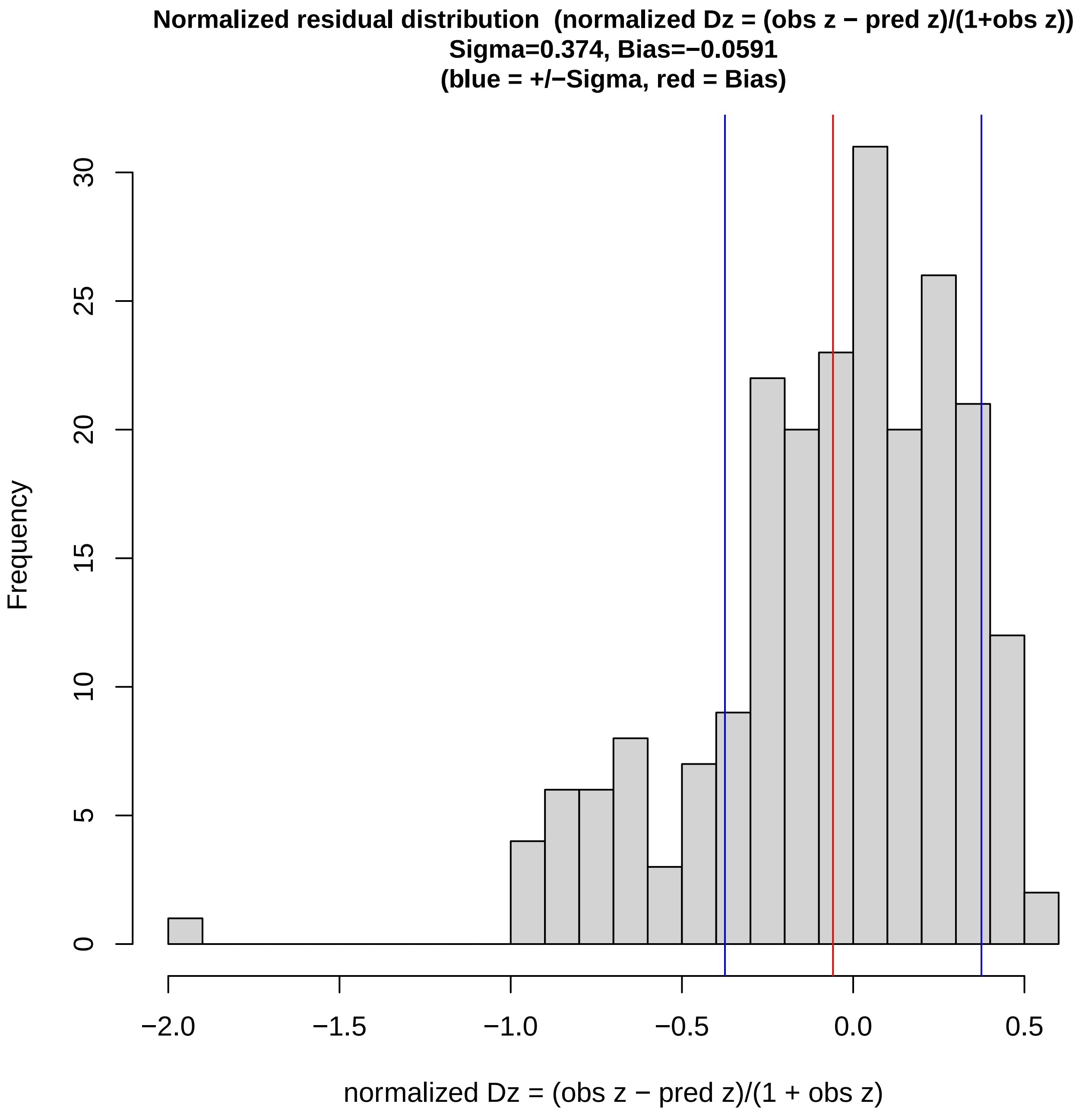


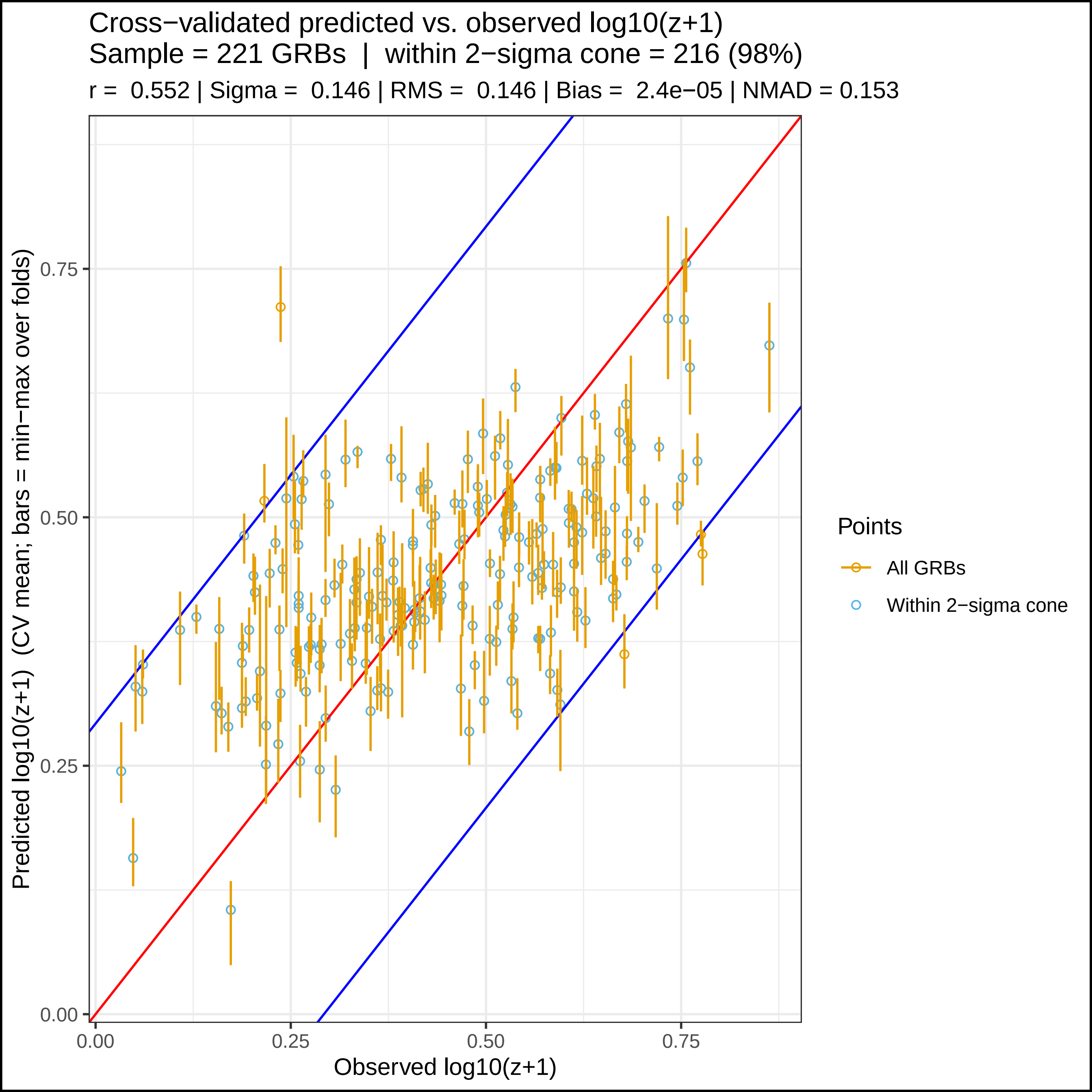
Redshift vs Normalized Bias
Normalized Bias
z
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
1
2
3
4
5
6


0.4
0.2
0.0
−0.2

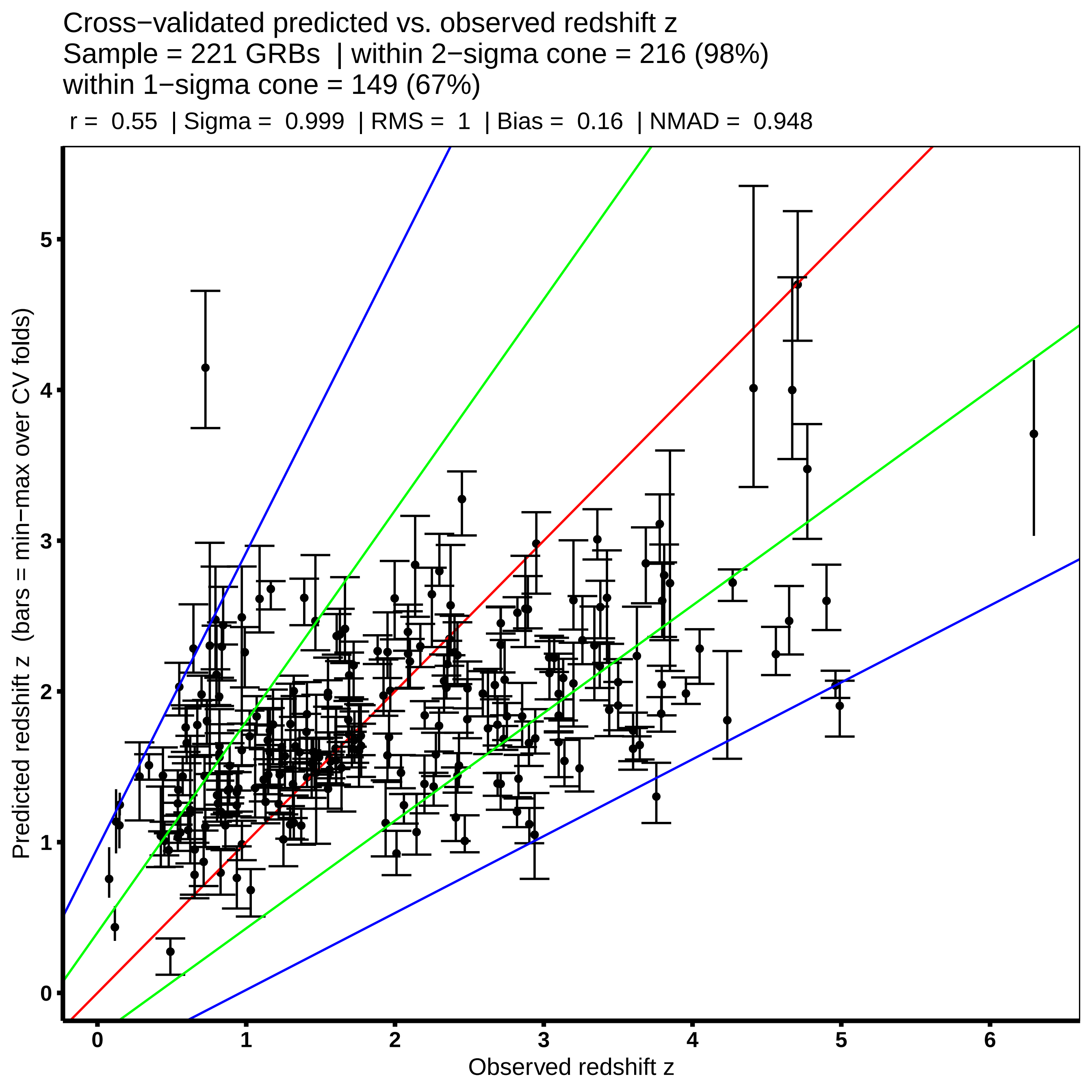


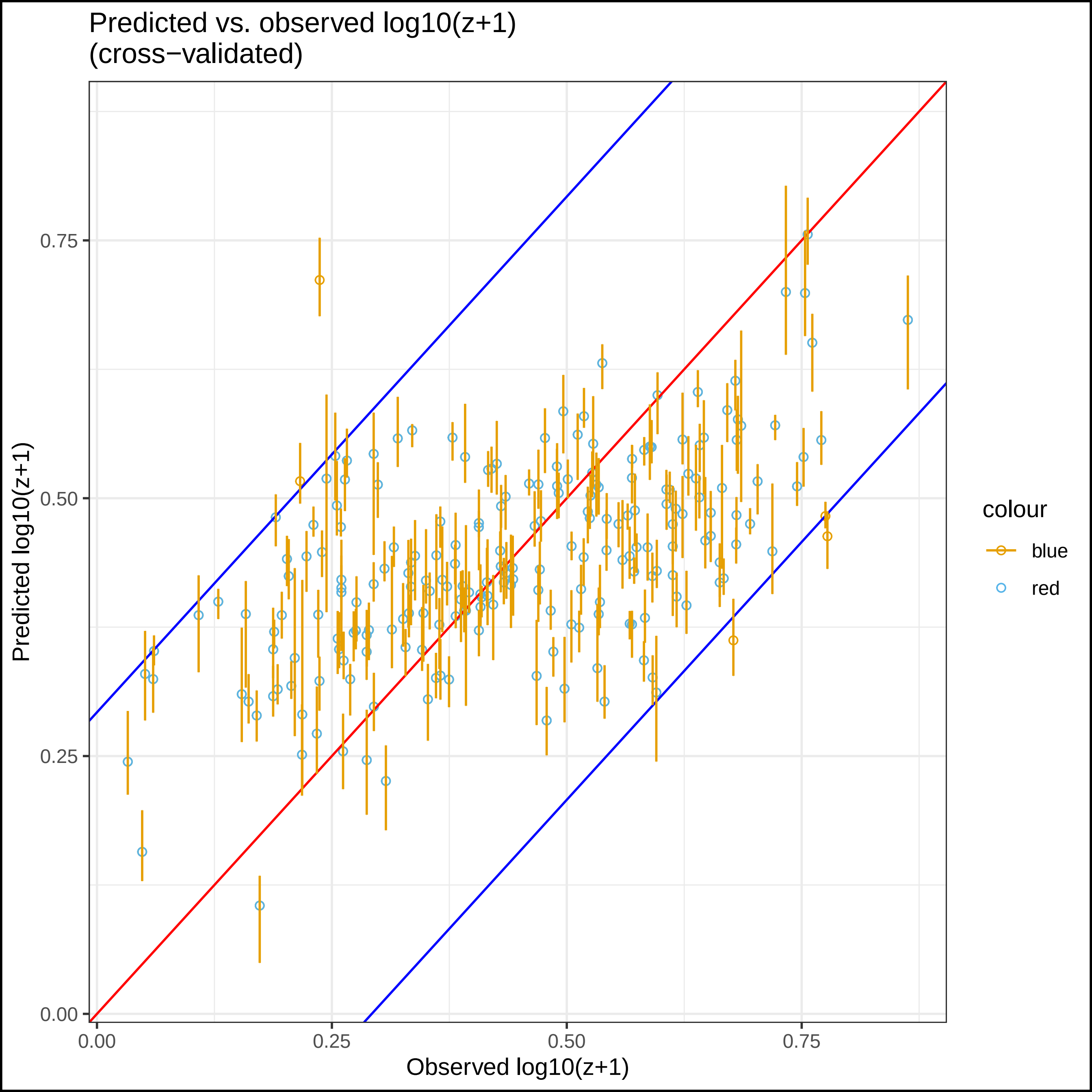
Redshift residual distribution (Dz = obs z − pred z)
Sigma=0.999, Bias=0.159
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
−3
−2
−1
0
1
2
3
Dz = observed z − predicted z

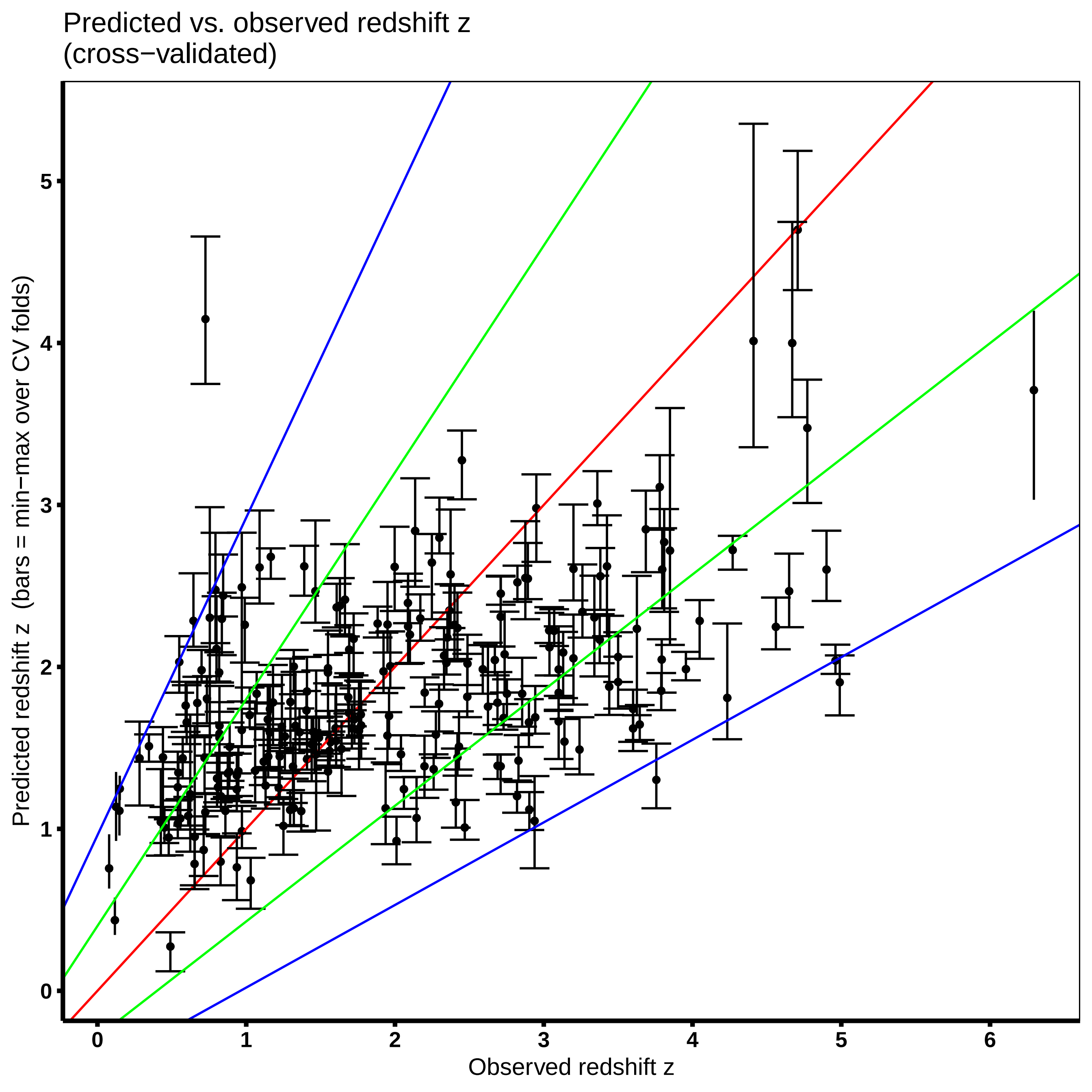

Normalized residual distribution (normalized Dz = (obs z − pred z)/(1+obs z))
Sigma=0.374, Bias=−0.0591
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
20
25
30
−2.0
−1.5
−1.0
−0.5
0.0
0.5
normalized Dz = (obs z − pred z)/(1 + obs z)

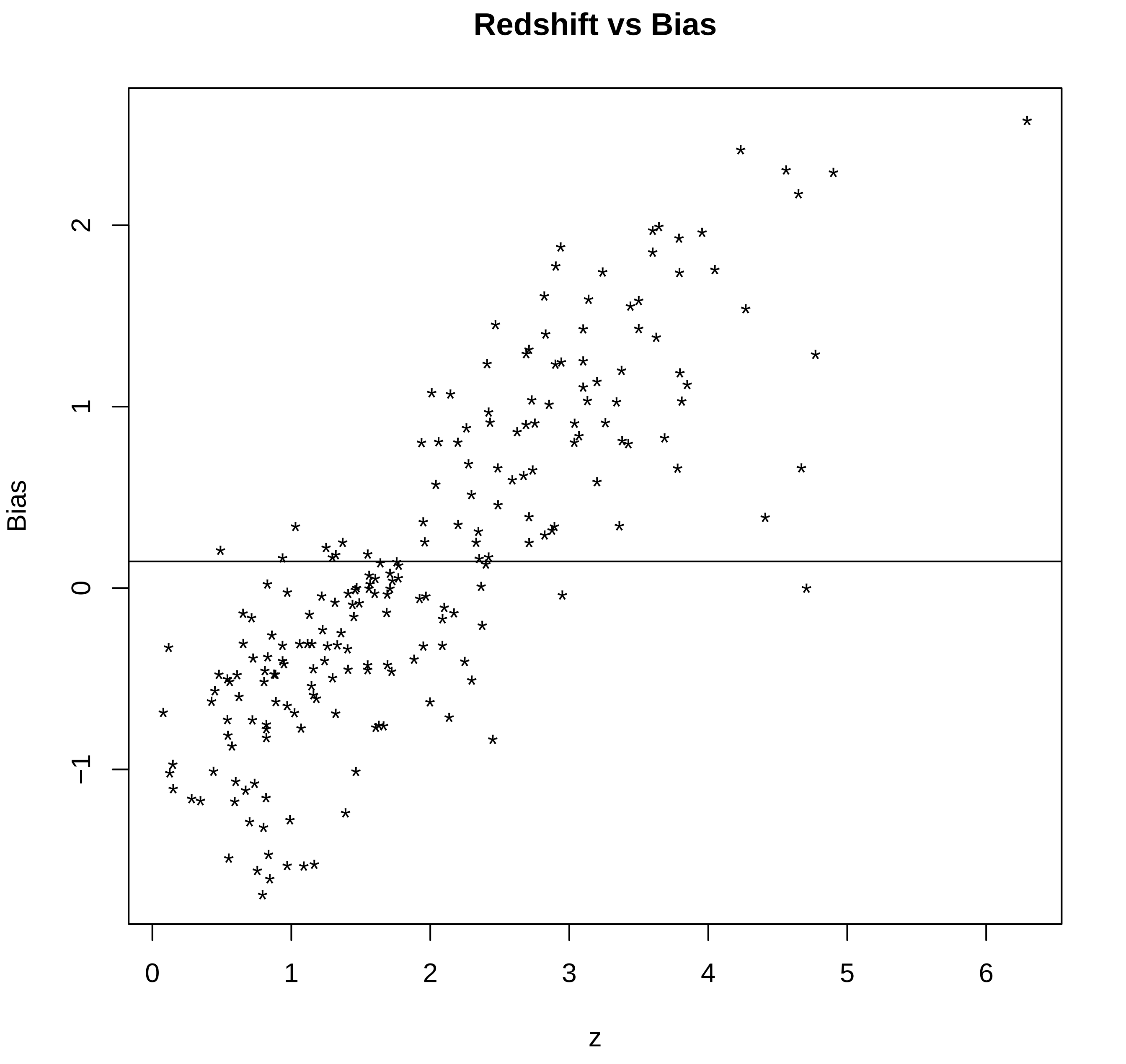


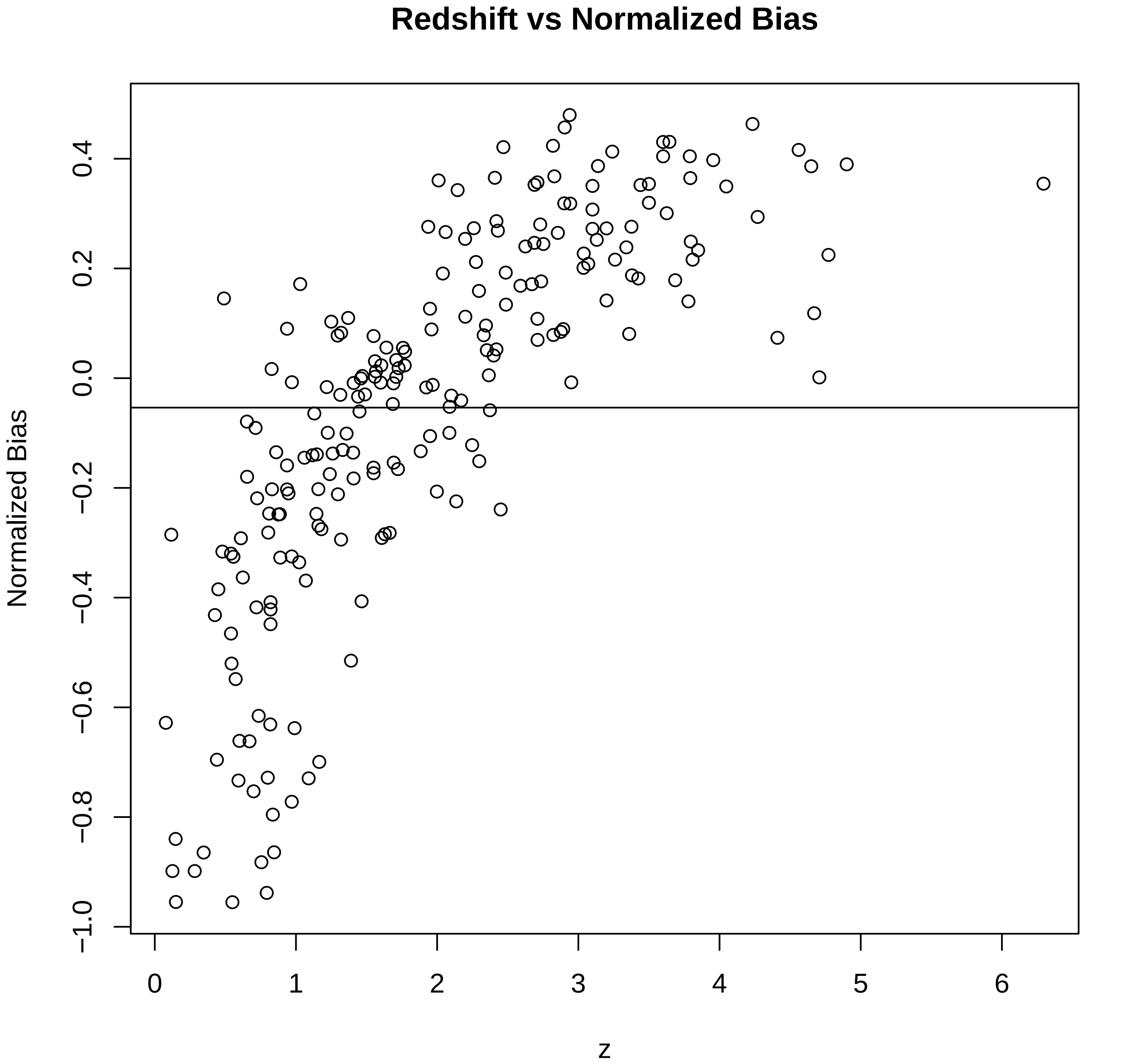
Cross−validated predicted vs. observed log10(z+1)
Sample = 221 GRBs | within 2−sigma cone = 216 (98%)
r = 0.552 | Sigma = 0.146 | RMS = 0.146 | Bias = 2.4e−05 | NMAD = 0.153
Predicted log10(z+1) (CV mean; bars = min−max over folds)
Observed log10(z+1)
0.00
0.25
0.50
0.75
Points
All GRBs
Within 2−sigma cone

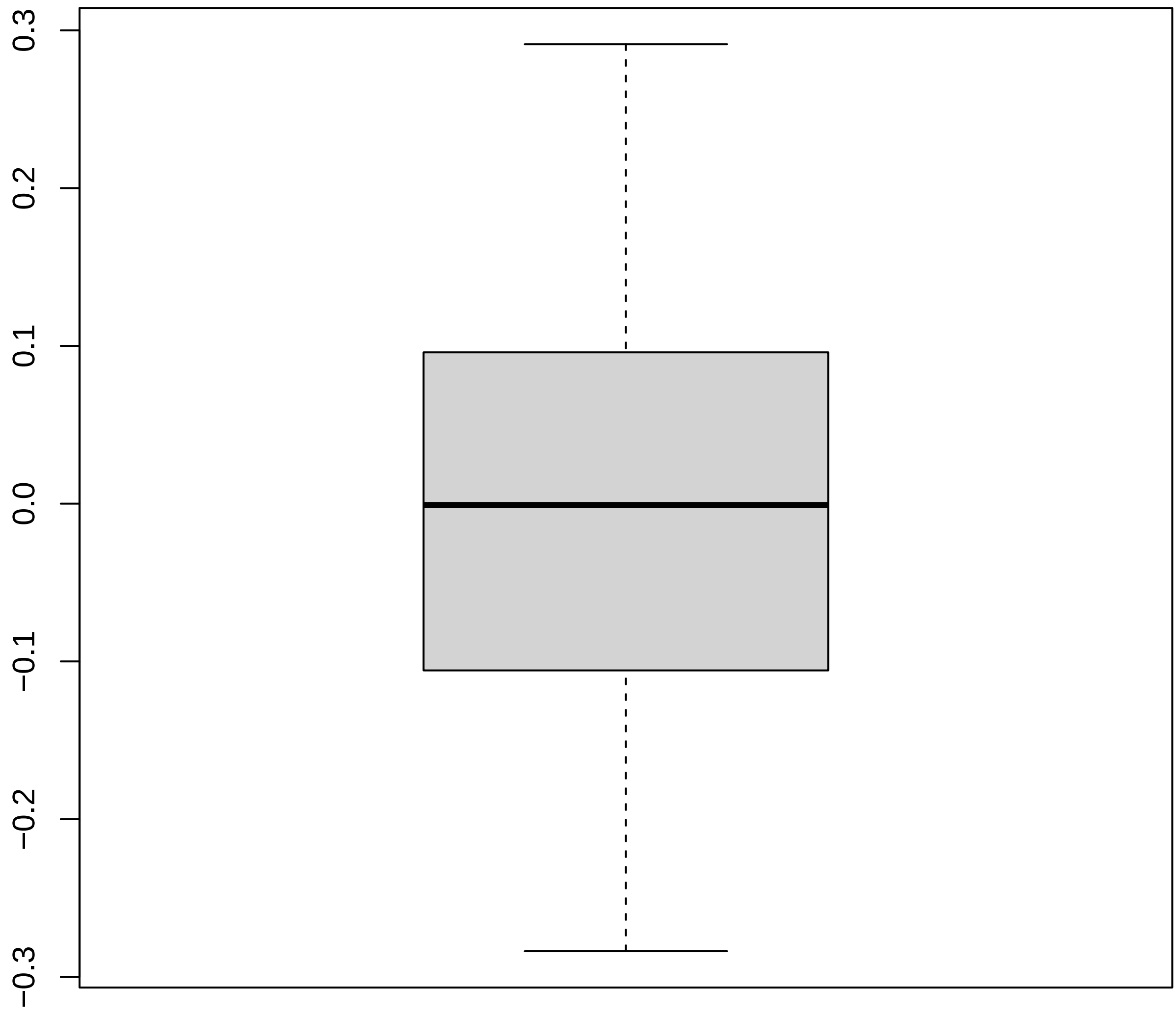

Cross−validated predicted vs. observed redshift z
Sample = 221 GRBs | within 2−sigma cone = 216 (98%)
within 1−sigma cone = 149 (67%)
r = 0.55 | Sigma = 0.999 | RMS = 1 | Bias = 0.16 | NMAD = 0.948
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6

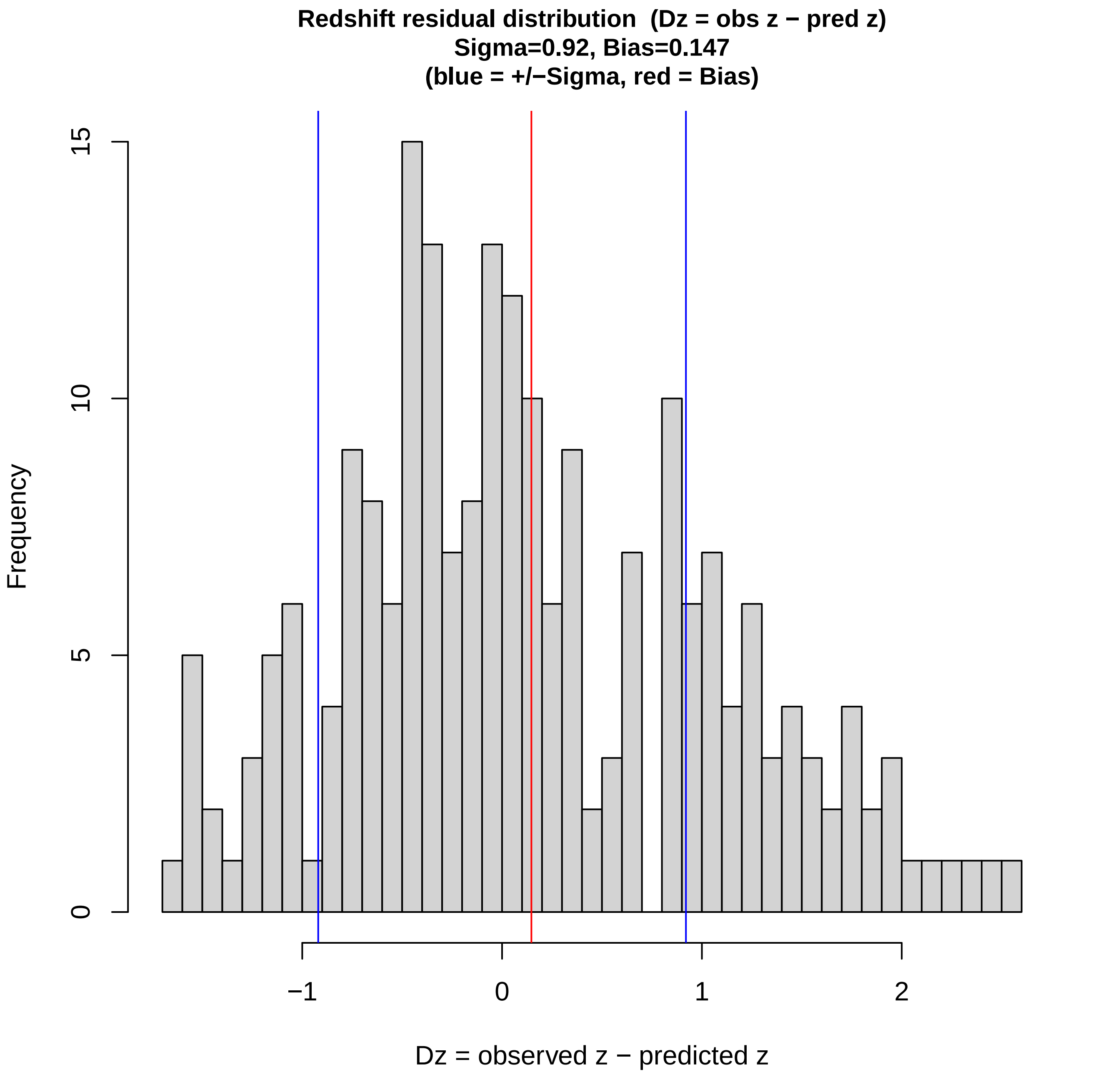

Predicted vs. observed log10(z+1)
(cross−validated)
Predicted log10(z+1)
Observed log10(z+1)
0.00
0.25
0.50
0.75
colour
blue
red


Predicted vs. observed redshift z
(cross−validated)
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6

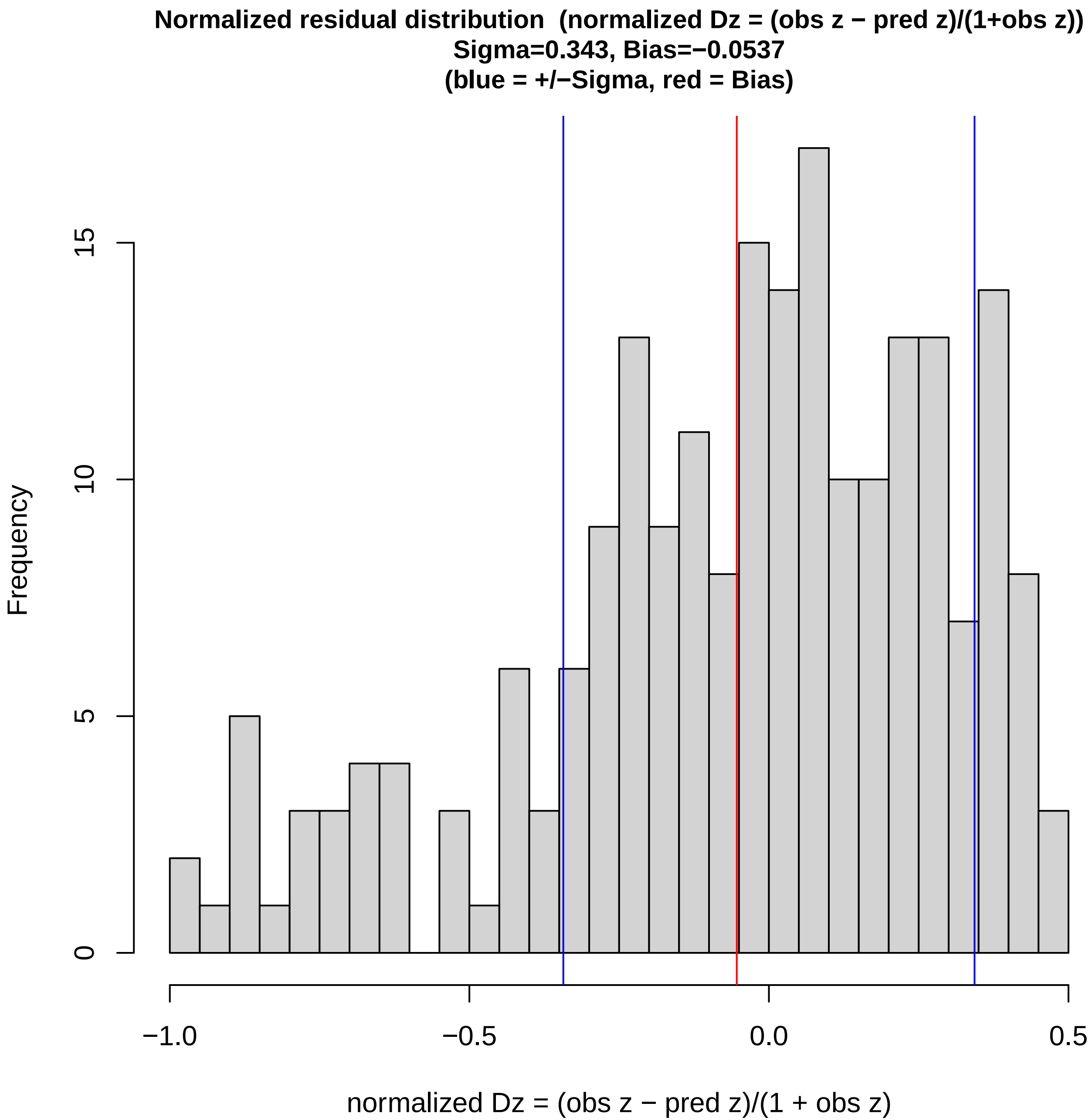

Redshift vs Bias
Bias
z
0
1
2
3
4
5
6
−1
0
1
2

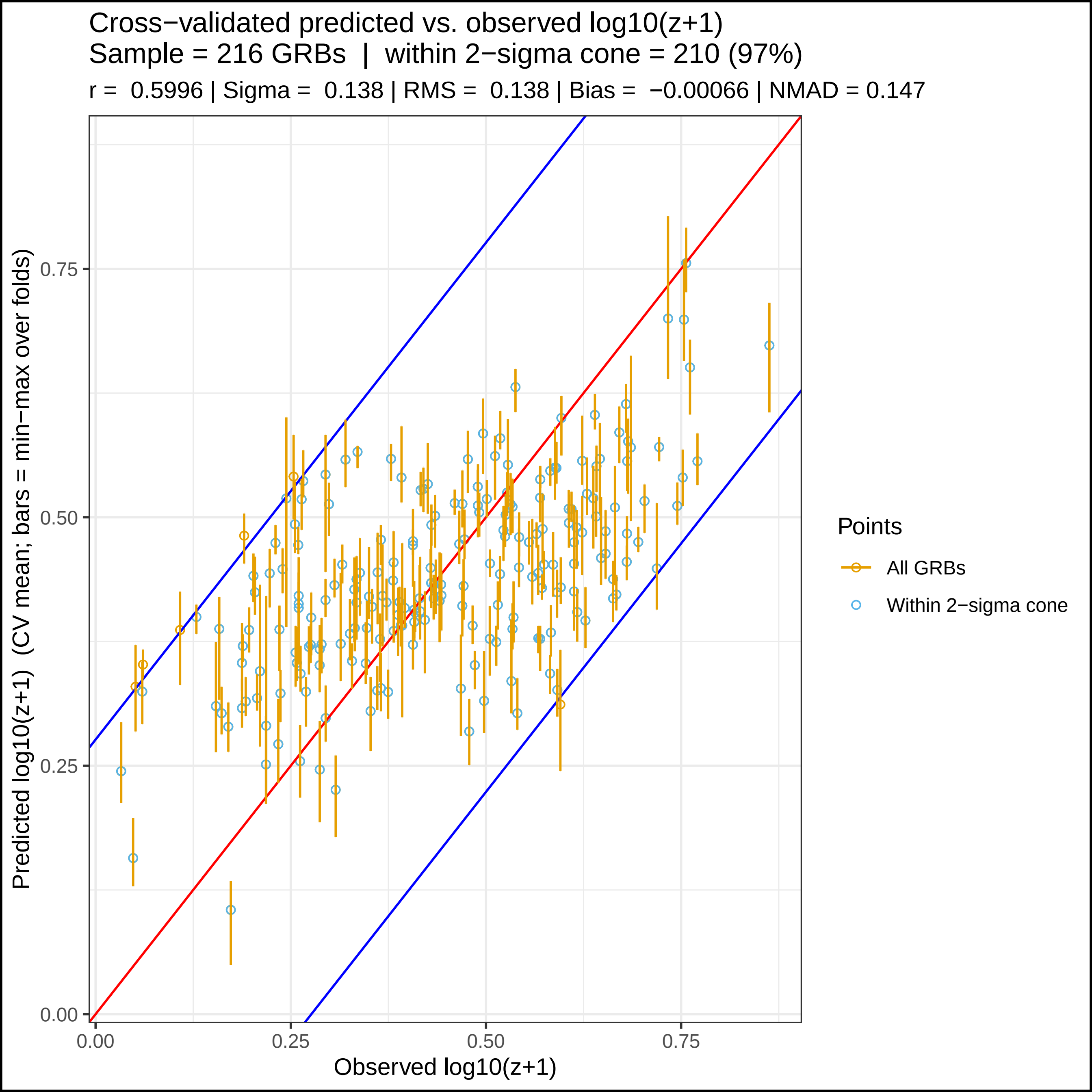

Redshift vs Normalized Bias
Normalized Bias
z
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
1
2
3
4
5
6

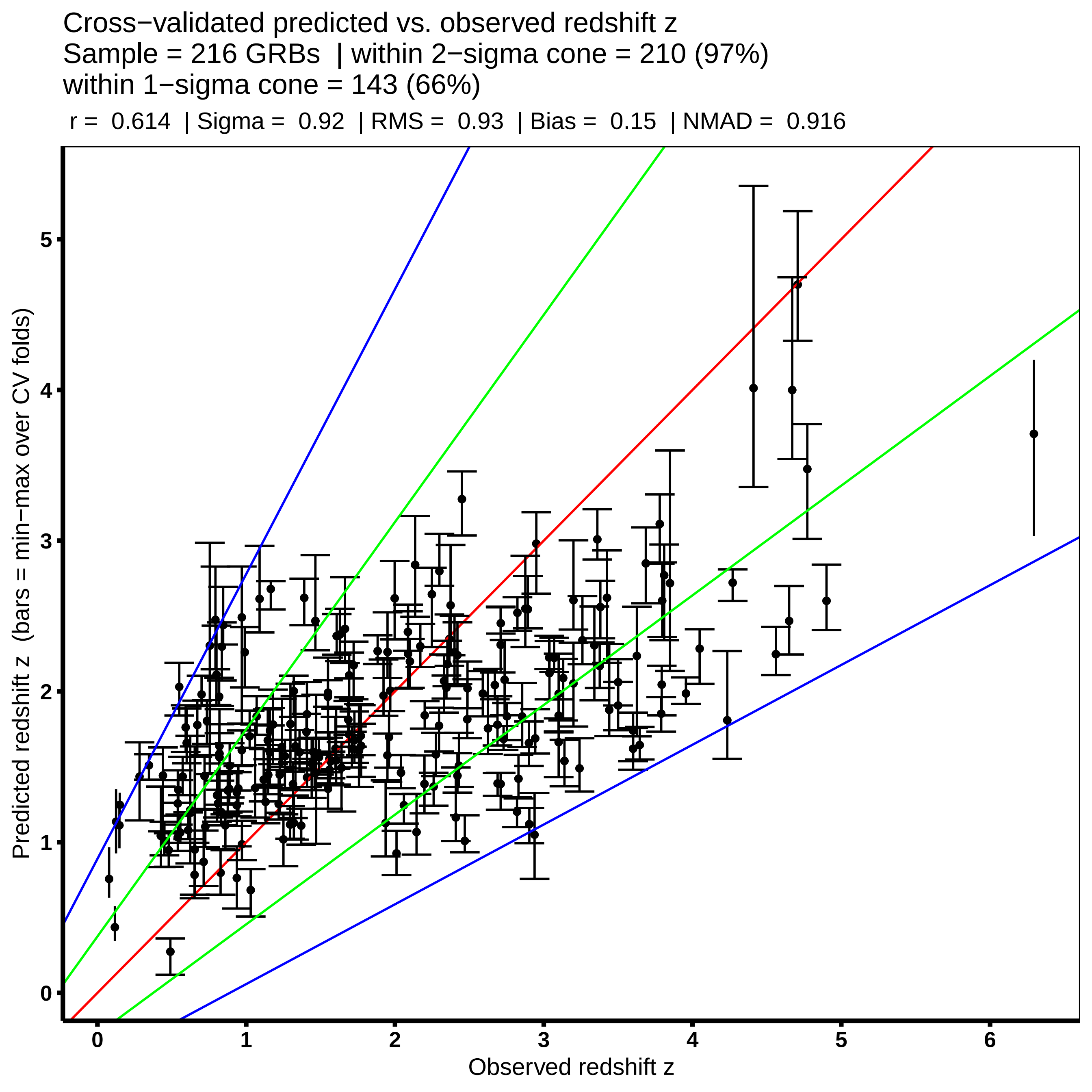

0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3

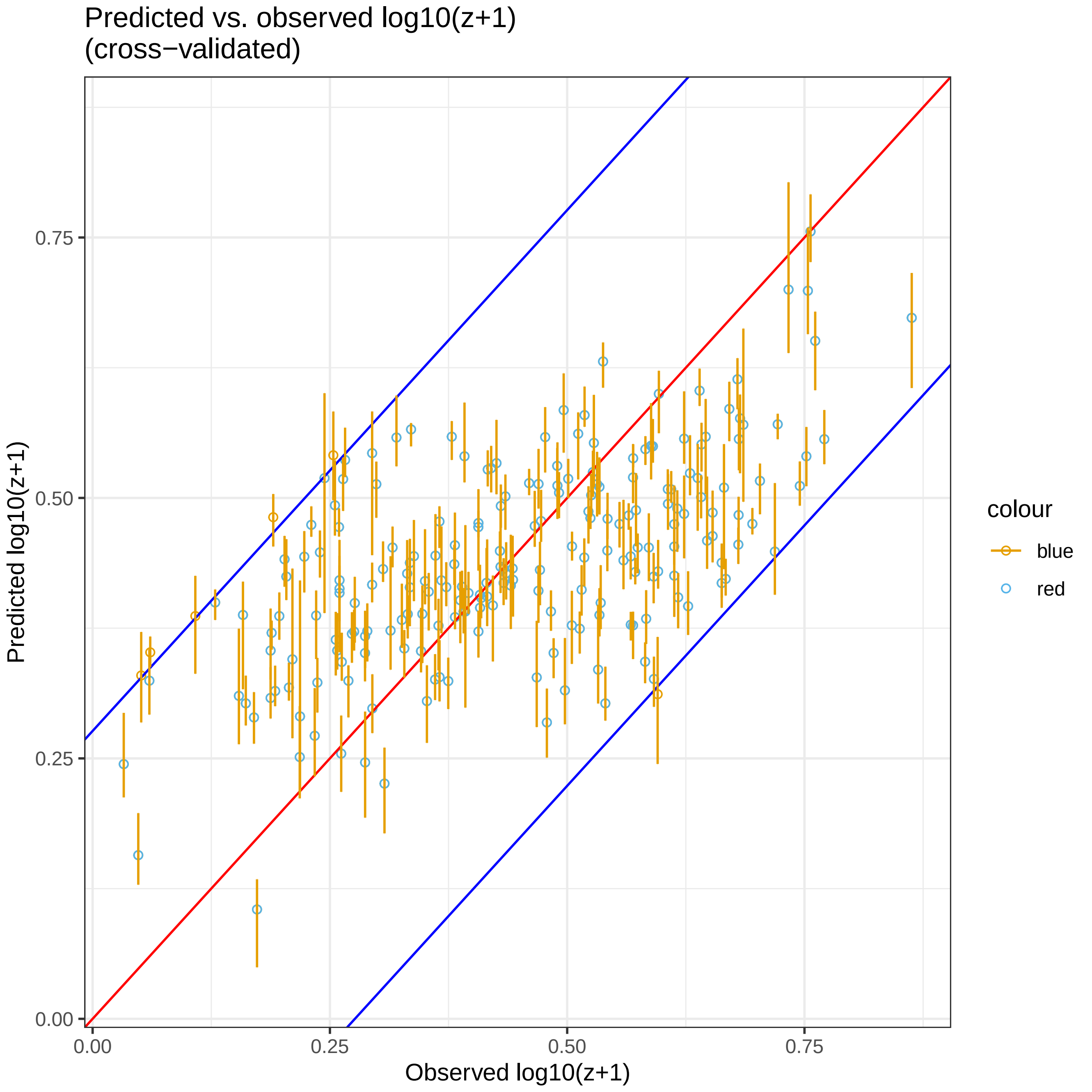


Redshift residual distribution (Dz = obs z − pred z)
Sigma=0.92, Bias=0.147
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
−1
0
1
2
Dz = observed z − predicted z

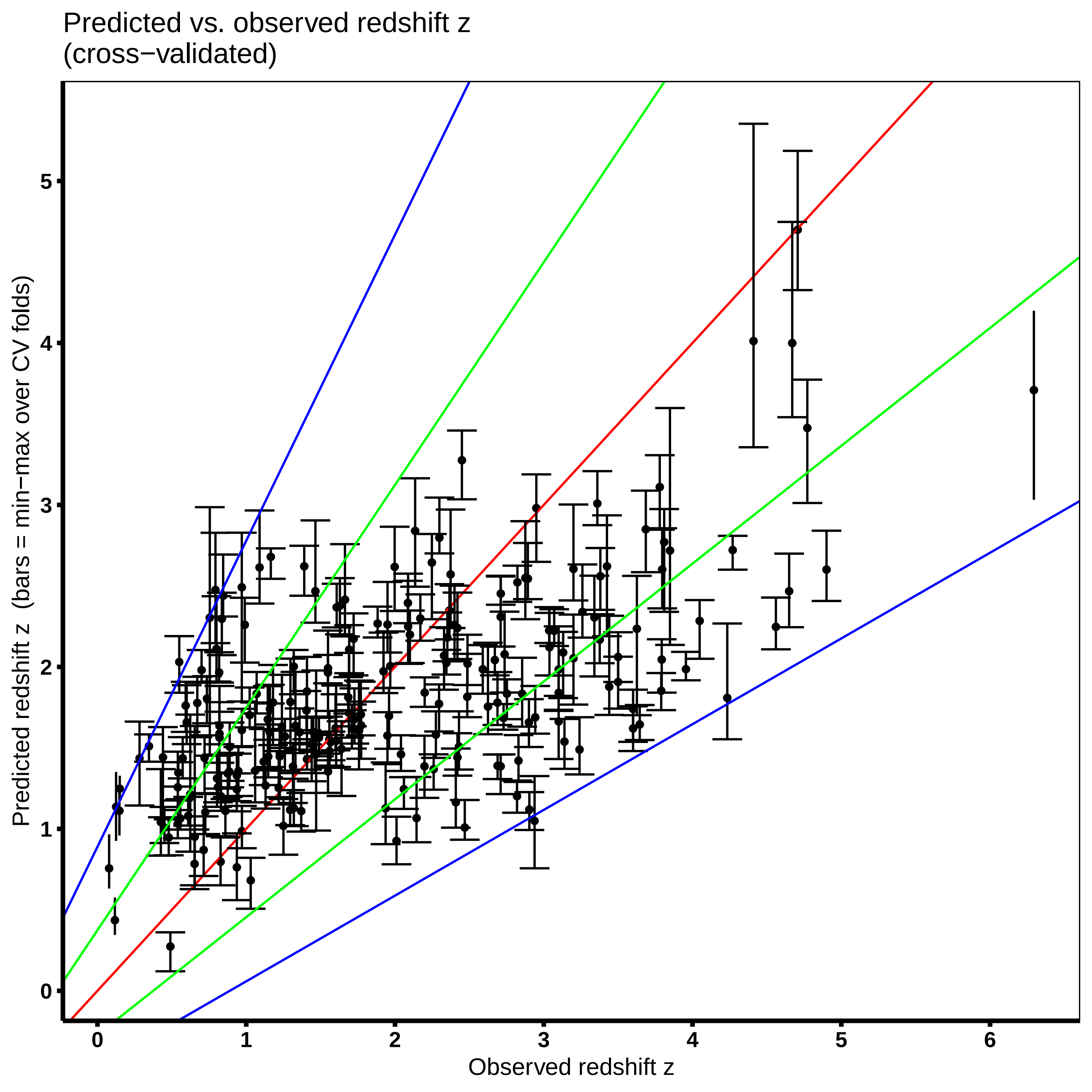

Normalized residual distribution (normalized Dz = (obs z − pred z)/(1+obs z))
Sigma=0.343, Bias=−0.0537
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
−1.0
−0.5
0.0
0.5
normalized Dz = (obs z − pred z)/(1 + obs z)


Cross−validated predicted vs. observed log10(z+1)
Sample = 216 GRBs | within 2−sigma cone = 210 (97%)
r = 0.5996 | Sigma = 0.138 | RMS = 0.138 | Bias = −0.00066 | NMAD = 0.147
Predicted log10(z+1) (CV mean; bars = min−max over folds)
Observed log10(z+1)
0.00
0.25
0.50
0.75
Points
All GRBs
Within 2−sigma cone


Cross−validated predicted vs. observed redshift z
Sample = 216 GRBs | within 2−sigma cone = 210 (97%)
within 1−sigma cone = 143 (66%)
r = 0.614 | Sigma = 0.92 | RMS = 0.93 | Bias = 0.15 | NMAD = 0.916
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6



Predicted vs. observed log10(z+1)
(cross−validated)
Predicted log10(z+1)
Observed log10(z+1)
0.00
0.25
0.50
0.75
colour
blue
red



Predicted vs. observed redshift z
(cross−validated)
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6